

Introduction and Overview

- **Significance:** The Indus Valley Civilization (IVC) and Gandhara Civilization are central to understanding South Asia's history and heritage.
- **Timeline:** It is one of the oldest and most advanced civilizations, flourishing between 3300 and 1300 BC.
- **Global Context:** It is considered a "cradle of civilization" alongside Mesopotamia, Egypt, and ancient China.
- **Location:** Primarily located in the northwestern Indian subcontinent, covering modern-day Pakistan and parts of India and Afghanistan.
- **Identity:** The IVC is a source of pride and identity for contemporary Pakistan.

Geography and The River System

- **Dependence:** Like the Egyptians with the Nile, the IVC relied heavily on the Indus River system.
- **Utility:** The river provided water for cleaning, a passage for trade, and nourishment for agriculture.
- **Extent:** The civilization stretched from the Himalayas (north) to the Arabian Sea (south), and from the Baluchistan Mountains (west) to the Thar Desert (east).
- **Size:** The area covered over 900 miles.
- **Rivers:** The Indus River is supported by five major tributaries: Jhelum, Chenab, Ravi, Beas, and Sutlej.
- **Hakra River:** Now a dry channel, it is visible in the Cholistan Desert.

Discovery and Major Cities

- **Excavation:** The civilization was undiscovered until the early 1900s; excavations by Sir John Marshall in 1921-1922 revealed complex urban settlements.
- **Harappa:**
 - Located in the Punjab region, about 100 miles southwest of Lahore.
- **Mohenjo-Daro:**
 - Located in the Larkana District of the Sindh province.
 - Known for its elaborate architecture and modern sewerage system.

- **City Structure:** Both cities featured "citadels" (heavily fortified central areas) and were situated along the Indus River to allow views into the distance.

Innovation, Infrastructure, and Technology

- **Standardization:** The people achieved great accuracy in measuring length and mass, being among the first to develop a system of standardized weights.
- **Bricks:** They used fire-baked, moisture-resistant bricks of uniform size, suggesting unity across the civilization.
- **Sanitation:** The IVC had the world's first known urban sanitation systems.
 - Systems were more advanced than contemporary sites in the Middle East and some modern areas of Pakistan and India.
 - Wastewater was directed from homes to covered drains on main streets.
 - Cities had private wells, toilets, and washrooms connected to street gutters.

Housing and Social Structure

- **Layout:** People lived in parallel rows of houses.
- **Class Division:** House size indicated economic status.
 - **Poor:** Lived in double-room houses.
 - **Rich:** Inhabited larger houses with more than two rooms.

Artifacts: Seals

- **Carving:** Harappans were known for seal carving—cutting patterns into the bottom face of small objects.
- **Usage:** Seals were used to identify property and stamp clay on trade goods.
- **Designs:** Commonly featured animal figures like elephants, tigers, and water buffalos.

Economy and Trade

- **Transport:**
 - Likely the first to use wheeled transport (bullock carts).
 - Built boats and watercraft, evidenced by a docking facility at Lothal.
 - Developed plank watercraft with sails for long-distance sea trade.

- **Trade Network:**
 - Extensive maritime trade existed between Harappan and Mesopotamian civilizations.
 - Harappan seals have been found in Iraq, Kuwait, and Syria.
- **Imports:** Minerals (Iran/Afghanistan), lead/copper (India), jade (China), and cedar wood (Himalayas).
- **Exports/Goods:** Terracotta pots, gold, silver, beads, pearls, and gemstones like lapis lazuli and turquoise.

Theories of Decline (The "Loss" Mystery)

Archaeologists have proposed several theories regarding why this organized society turned into ruin:

- **1. Floods:** Massive floods may have destroyed the cities; Mohenjo-Daro was rebuilt seven times.
- **2. Droughts/Climate Change:** Evidence suggests severe drought 4,000-3,500 years ago, changing monsoon patterns, and altered river courses disrupted agriculture.
- **3. Foreign Invasion:** Unburied skeletal remains found in Mohenjo-Daro suggest genocide or invasion.
- **4. Epidemic:** Bones showing "painful struggle" suggest a plague, though proof is limited.
- **5. Tectonic Disturbances:** Some geologists believe a major earthquake caused the decline.
- **6. Nuclear Attack Theory:**
 - Some argue a sudden "nuclear" type event occurred because death came quickly.
 - Evidence includes an epicenter where terrain appears crystallized/melted (requiring heat over 1500°C) and bricks fused facing away from the center.
- **7. Socio-Political Dynamics:** Regional conflicts may have disrupted trade, leading to social unrest and the abandonment of urban centers.

Transition to Vedic Era

- **Shift:** The decline led to a transition from a highly organized urban civilization to a decentralized, rural, and pastoral society.
- **Culture:** This era introduced new religious rituals, social structures, and linguistic developments found in Vedic texts.

Conservation and Modern Significance

- **Deterioration:** 30% of Mohenjo-Daro's ruins have collapsed since the 1920s.
- **Urgency:** Pakistan is racing against time to protect the site before it disappears due to deterioration.
- **Heritage:** The IVC shapes Pakistan's national identity, with motifs appearing in art and textiles.
- **UNESCO:** Harappa and Mohenjo-Daro are UNESCO World Heritage Sites.
- **Future:** Preserving this link to the past fosters national pride and offers lessons in sustainable development and planning.